Soumyajyoti Biswas

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Summary

Software Engineer with production experience designing, debugging, and operating **backend microservices under zero-trust controls** (Python, FastAPI, Java, Spring Boot, PostgreSQL, Docker). Ran services against a **99.9% availability SLA** and shortened incident resolution time through log-driven root-cause analysis; test- and documentation-heavy by default. Also ships **LLM applications** end-to-end.

Experience

Independent Software Engineer *Bangalore, India*
Self-Directed Projects *Jan 2026 - Present*

- **GNSS Interference Detection Pipeline** – Python/DuckDB ingestion and analytics pipeline over live ADS-B telemetry across **12 regions** (29k observations, 16 SQL views), with a crash-safe disk buffer and idempotent MERGE loads; replaced global-threshold detection with per-airframe statistical baselining, cutting false positives **67% to 0%** on synthetic fixtures. **75 unit tests, 128 integration checks**, ADRs and an operational runbook.
- **Vera Engagement Bot** (magicpin AI Challenge) – stateless FastAPI service over a 26-kind trigger taxonomy; cut /v1/tick latency **54s to 0.07s** via precompute-on-ingest, then replaced the LLM critique call with a typed-decision model (TypeSafe Jev) behind a swappable engine interface with regex fallback. **130 pytest tests**.

Software Engineer *Kochi, India*
IBM Labs – WatsonX Governance *Mar 2024 - July 2025*

- Owned **drift-detection microservices** in the WatsonX OpenScale squad implementing **Kolmogorov–Smirnov**, **Population Stability Index**, **Jensen–Shannon divergence**, and Wasserstein-distance tests to flag data, model, and prediction drift across enterprise customer ML pipelines – restructured as **Spark batch jobs** over precisely bounded windows to cut evaluation runtime on terabyte-scale datasets.
- Maintained containerized **RAG backend services** under strict zero-trust controls (audit logging, entitlement checks, rotated service tokens) against a **99.9% availability SLA** across high-throughput distributed environments.
- Led **root-cause analysis and debugging** of complex live-site incidents by correlating logs across distributed services, shortening time-to-resolution and eliminating recurring failure classes while resolving customer-reported issues and security concerns ahead of release deadlines.
- Enhanced governance microservices that trigger **automated retraining workflows** on detected drift, improving alert signal-to-noise for downstream operators.
- Architected a **dual-timestamp mechanism** (scoring vs storage) over terabytes of historical and real-time data, enabling reproducible model evaluation across precise, time-bounded windows.
- Built cross-database **DDL utilities and validators** (DB2, PostgreSQL) and end-to-end Hive notebook workflows for dataset creation, payload validation, and analytics across enterprise customer environments.

Software Engineer Intern *Bangalore, India*
Nineleaps *Feb 2023 - Oct 2023*

- Co-built the **Java / Spring Boot** backend of a fundraising platform (15 REST controllers, 16 JPA entities) with JWT auth, OTP onboarding, and role-based access control for 4 user types.
- Built the bidding/investment-window engine on **PostgreSQL** (scheduled runners, ACID transactions); authored **software tests to 88%+ coverage** with GitHub Actions CI/CD.

Projects

CogniStream AI - Autonomous Video Pipeline FastAPI · LangGraph · ChromaDB

- Built a 3-service FastAPI + LangGraph stack orchestrating ComfyUI/Wav2Lip/ESRGAN on a single RTX 4070, with a self-learning ChromaDB retrieval layer (auto-tunes on OOM) and an OpenTelemetry-traced Redis GPU lock for slot-based parallelism.

Finance Prediction System - Multi-Asset Forecasting XGBoost · HMM · Streamlit

- Built a multi-asset forecasting system (30 US large caps, 20 NSE tickers, crypto/FX) on XGBoost with isotonic calibration – 65-72% top-decile accuracy at Brier 0.22, validated with purged k-fold and leakage checks; added a 4-state HMM regime engine and a Streamlit dashboard.

Technical Skills

Languages & Backend: Python, Java, SQL, FastAPI, Spring Boot, REST APIs, Microservices, PostgreSQL, DB2, DuckDB, Redis, Docker
Testing & Tooling: Pytest, GitHub Actions CI/CD, OpenTelemetry, Git, Postman, technical documentation & runbooks
Data & ML: Spark, pandas, NumPy, Scikit-learn, XGBoost, PyTorch, Drift Detection, Statistical Validation
GenAI & LLM: LLMs, RAG, Agentic Systems, LangGraph/LangChain, Hugging Face, Transformers, Prompt Engineering, Copilot-assisted development

Education

Acharya Institute of Technology - B.E. Computer Science, 7.7 CGPA *May 2019 - May 2023*